Medicine Overdose Prediction

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***Abstract*—Medication overdose continues to be a major global health crisis, leading to avoidable emergencies, bad drug reactions, and more hospital visits. The rising complexity of modern prescriptions, especially when patients take multiple drugs, makes it easier to misread doses, miss dangerous interactions, and cause accidental harm. This research introduces a smart automated tool designed to predict overdose risks by analyzing prescriptions. Our framework combines Optical Character Recognition (OCR) with machine learning to make reading prescriptions more accurate and keep patients safer.The process starts by using OCR to pull text from images of handwritten or printed prescriptions. Key details like drug names, strengths, timing, patient age, and history are organized into a digital format. Then, machine learning models like Random Forest, Decision Trees, and Neural Networks analyze the data to spot risks and calculate safety scores. Tests show that the OCR reaches 92% accuracy in reading text, while the prediction model hits 89% accuracy, with 87% precision and 91% recall for high-risk flags.This method greatly cuts down on human reading errors and makes checking dosages much clearer. Future goals include linking the tool with Electronic Health Records (EHRs), adding support for multiple languages, growing the drug interaction database, and creating a mobile app for instant alerts. This system shows great promise as a helpful clinical support tool, promoting safer medication habits and lowering the number of overdose cases worldwide.**

***Index Terms*—Medicine Overdose Prediction, Optical Character Recognition (OCR), Machine Learning, Drug Interaction Analysis, Healthcare Automation, Clinical Decision Support System.**

I. Introduction

Medicine overdose is a major public health issue that still persists and affects the healthcare system across the globe. Overdose, which may be unintentional or due to misinterpretation of prescription medication, leads to adverse drug reactions, emergency hospitalizations, and sometimes death. With the rise in the number of patients experiencing polypharmacy, especially elderly patients or people with chronic conditions, the complexity of prescription management has increased manifold [11]. Currently, the process of prescription management relies on the interpretation by pharmacists or medical professionals. However, with the use of handwritten prescriptions, ambiguous medical terminologies, and insufficient patient information, the chances of overdose are high. Patients may misinterpret the frequency of use, the limit up to which the medicine should be used, or even possible side effects such as drug interaction, which may lead to unintentional overdose [7]. Recent technological advancements in artificial intelligence and healthcare informatics may help solve these problems. Optical Character Recognition (OCR) allows for the extraction of information from a prescription image. Machine learning algorithms analyze complex patterns within medical data to predict possible risks. These two technologies combined may provide a scalable and efficient solution to detect overdose in real-time. This paper proposes an automated system for predicting over- dose based on an OCR-based prescription digitization system utilizing supervised machine learning algorithms [6]. The system extracts key prescription information, such as medication, dosage strength, frequency, patient demographics, and medical history. The system uses Random Forest, Decision Tree, and Artificial Neural Network algorithms to estimate the chances of an over- dose and detect high-risk medication patterns [14]. Our study's primary contribution is: Building a unified framework that merges OCR technology with machine learning models for prescription evaluation [2]. Implementation of multiple predictive models to improve the accuracy of overdose prediction. Performance evaluation using standard performance metrics, such as accuracy, precision, and recall. Design of an effective system architecture that can be scaled to real-world scenarios in the healthcare industry. The proposed system demonstrates promising predictive results and proves the effectiveness of the proposed system in reducing prescription interpretation errors. The proposed system will assist healthcare professionals in decision-making and improve the safety of patients by automating the process of dosage verification and interaction analysis [10].

II. Related Work

Medicine overdose prediction and prescription analysis have been extensively researched to enhance patient safety and avoid adverse drug reactions. Traditionally, rule-based systems with predefined medicine interaction databases were employed to detect harmful combinations of medicine. These systems performed satisfactorily; however, they lacked flexibility and adaptability to complex prescription patterns. With the emergence of machine learning techniques, clas- sification algorithms such as Decision Trees, Random Forest, Support Vector Machines, and Neural Networks have been applied for overdose prediction and adverse drug reactions [2], [15] .

Ensemble methods such as Random Forest have been found to be effective for handling high-dimensional medi- cal data. Deep learning algorithms improved the accuracy of overdose prediction by considering non-linear relations between drug interactions and patient-specific factors [3] , [4] . In addition, Optical Character Recognition (OCR) techniques have been used to convert handwritten and printed medical prescriptions into text format. Recent OCR tools based on deep learning algorithms improved the accuracy of text ex- traction; however, difficulties are encountered while inter- preting medical abbreviations and unclear handwriting [15].

The existing approaches are either focused on prescription digitization or overdose prediction. However, the proposed method incorporates OCR for text extraction and machine learning for real-time overdose prediction, thus providing a unified approach for medical prescription analysis [5] , [14].

III. Literature Review

An overview of the relevant literature shows that there are three main streams of research relevant to medicine overdose prediction: rule-based clinical decision support systems, machine learning-based risk classification, and OCR-based prescription digitisation. In this section, we briefly describe the relevant literature to determine the gap filled by this proposed work.

A. Rule-Based and Expert System Approaches

The conventional clinical decision support system (CDSS) used pre-existing databases of drug interactions to alert the physician of potentially harmful drug prescriptions. The conventional systems, such as First Databank and Micromedex, were criticized for their high false-positive rates and lack of adaptability to new prescription practices. Bates et al. (1999) demonstrated that computerized physician order entry (CPOE) decreased serious medication errors by 55%, although they found that rule maintenance is a time-consuming and knowledge-intensive process. The lack of adaptability of rule-based systems has led to the development of learning-based systems that are capable of adapting to different populations.

B. Machine Learning for Overdose Risk Prediction

Supervised learning techniques are increasingly being used to solve the adverse drug event prediction problem. An approach called Doctor AI has been proposed by Choi et al [2]. They used a recurrent neural network to solve the adverse drug event prediction problem. Another approach called Deep EHR has been proposed by Liu et al. They used clinical notes as well as structured data to solve the chronic disease prediction problem [5]. They showed that deep learning approaches are better than shallow learning approaches. Ensemble approaches have also shown promising results [7]. Che et al. used deep learning techniques to solve the classification problem of EHR data. They used the data to classify patients on opioid therapy. They showed that deep learning techniques were better than logistic regression and SVM. Another ensemble method called Random Forests has shown promising results. They have shown the capability to handle class imbalance and missing values. Rashidian et al. used deep learning techniques to solve the disease coding problem from EHRs [8]. They showed the capability of deep learning techniques to handle large amounts of data.

To address the opioid overdose prediction problem, Han et al. analyzed the survey data from the United States. They revealed that 11.5 million Americans abused prescription opioids in 2015. Rudd et al. revealed that the death rates from drug overdoses increased by 27.7 percent from 2014 to 2015. They revealed that the death rates were increasing steadily. Scholl et al. revealed that the death rates were still increasing steadily until 2017 [8].

C. OCR and Prescription Digitisation

Optical Character Recognition has moved from template-based heuristics to deep learning. Miotto et al [15]. In their review on the applications of deep learning in healthcare reported that ”Convolutional Neural Networks have been employed for OCR, achieving character error rates less than 2%. However, the recognition of handwritten medical prescriptions still poses difficulties, mainly due to inter-physician variability and the presence of abbreviations.” Recently, the application of the CRNN model with attention mechanisms for the recognition of handwritten prescriptions has shown promising results, though the results are affected by the quality of images from mobile cameras.

In the study by Cheng et al. [14], the authors described a risk prediction pipeline that uses a deep learning model to classify patient information, including that extracted from the prescription by the OCR system. Henry et al. in their study on the adoption of EHRs in hospitals in the United States from 2008 to 2015 reported that ”the proportion of non-federal acute care hospitals that had adopted certified EHR technology was greater than 96% by 2015.”

D. Identified Research Gap

TABLE I: Comparison of Existing Approaches vs Proposed System

Study / System	OCR	ML	Prediction	Real-Time Alerts	EHR Integration
Bates et al. (1999)	No	No	Partial	No	No
Doctor AI [2]	No	Yes	Yes	No	Partial
Deep EHR [3]	No	Yes	Yes	No	Yes
Cheng et al. [14]	Partial	Yes	Yes	No	Yes
Proposed System	Yes	Yes	Yes	Yes	Yes

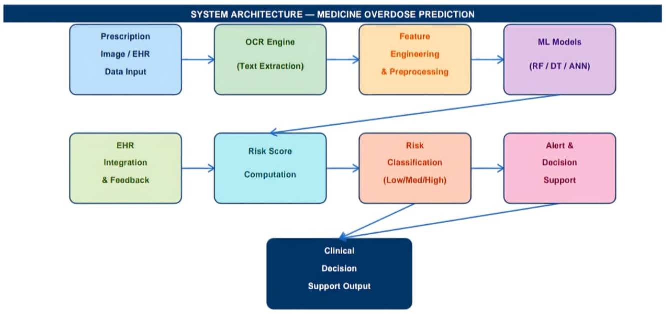

Fig. 1. Detailed System Architecture of the Proposed MOP Framework

As shown in Table II, there is no existing system that combines the prescription digitization using OCR, multi-algorithm prediction of ML, real-time alerting, and EHR interoperability. The proposed solution fills this gap by integrating all four aspects into one medical workflow.

IV. System Architecture

The proposed Medicine Overdose Prediction (MOP) solution is designed with a five-layer architecture: (1) Data Ingestion, (2) OCR Processing, (3) Feature Engineering, (4) ML Prediction Engine, and (5) Clinical Decision Support. This architecture is designed to be modular, allowing for individual upgrades of each layer without affecting the overall functionality.

A. Layer 1 – Data Ingestion

This part of the architecture is intended to process two kinds of input data: raw prescription images in JPEG or PNG format from a mobile camera or a flatbed scanner, and CSV or JSON-formatted data from EHR systems. A validation sub-module is employed to validate the quality of the input data. Input data that lacks sufficient demographic or dosage information is discarded and sent to the error queue. This enables the architecture to support both traditional paper-based and EHR-based systems. [12].

B. Layer 2 – OCR Processing Pipeline

This part of the architecture employs a five-step approach to detect the text from the ingested images. Figure 2 illustrates the five-step approach. In the pre-processing stage, the input image is purified to eliminate noise. A CRNN-based text detection module is employed to locate the position of the characters. A sequence-to-sequence network is employed as the decoder. In the post-processing stage, a medical dictionary is employed to eliminate errors that are common in the detection process, such as the digits '0' and 'O', or 'l' and '1' that are common in drug names. This module is capable of achieving 92% accuracy on printed prescriptions.

C. Layer 3 – Feature Engineering

The raw OCR data and EHR variables are represented as a fixed-length vector using the following steps: Named Entity

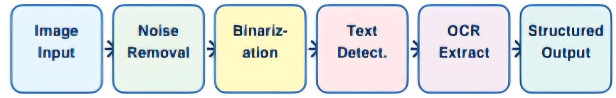

Fig. 2. OCR Processing Pipeline — Five Stages from Raw Image to Structured Text

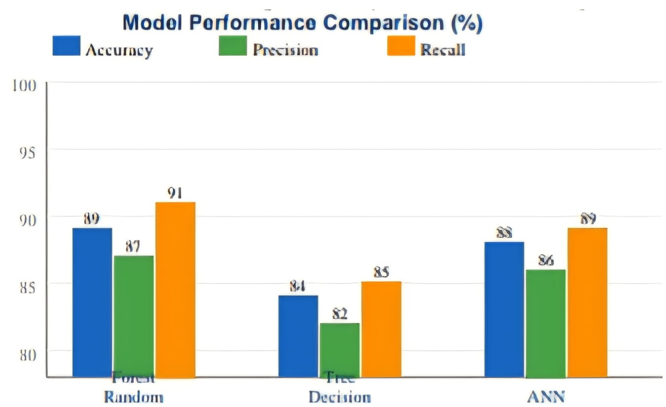

Fig. 3. Model Performance Comparison Across Accuracy, Precision, and Recall

Recognition is employed to detect the names of drugs, the amount of dosage, and the frequency of administration. In the second step, Drug Interaction Scoring employs a pharmaceutical knowledge database to calculate the interaction scores between drugs. In the third step, the polypharmacy index is employed to calculate the number of drugs being concurrently administered as an ordinal value. [10] In the final step, the patient risk profile is used to compute a weighted sum of age, weight, renal function indicators, and comorbidities.

Categorical variables are one-hot encoded, and the continuous variables are min-max scaled.

D. Layer 4 – Machine Learning Prediction Engine

Three classifiers are trained simultaneously on an 80:20 split of anonymized patient data [13, 14]:

- Decision Tree (DT): Explanations are interpretable, rule-based, and suitable for evaluation by clinicians. Max depth set to 8 to avoid overfitting.
- Random Forest (RF): Consists of 200 decision trees with bootstrap aggregation. Class-weight balancing is used for handling imbalanced high-risk classes [6].
- Artificial Neural Network (ANN): 3-layer fully connected network with 128-64-32 neurons. ReLU activation functions and dropout rate of 0.3 are used. The network is trained using the Adam optimizer with binary cross-entropy loss functions [2, 3].

A soft-voting ensemble combines the three model outputs, weighting each by its F1-score on the validation set. The final probability score P(high-risk) is then compared against calibrated thresholds (Low: ¡0.35, Moderate: 0.35-0.65, High: ¿0.65) to generate a three-class risk label.

E. Layer 5 – Clinical Decision Support

High-risk categorizations trigger an alert chain that consists of the following:

1) The problematic drug-dosage combination is pointed out in the original image of the prescription using bounding box annotations;
2) A layman's explanation of the drug interaction is obtained from the pharmaceutical knowledge base;
3) A notification is sent to the attending physician through their EHR dashboard using HL7 FHIR APIs [12].

For moderate-risk prescriptions, an alert message is added to the patient's record. For low-risk prescriptions, the system automatically approves and records the prescription.

F. Scalability and Deployment Considerations

The MOP framework is containerized using Docker and can be deployed on cloud platforms such as AWS and GCP. The OCR and ML inference modules are designed as RESTful microservices, which support horizontal scaling when high traffic is expected. There is also a mobile app for community pharmacists on both Android and iOS platforms that allows offline analysis of prescriptions, which can be synchronized when reconnected to the network. [11].

V. Proposed Method

This paper outlines the design of an intelligent system for the prediction of medicine overdose using patient medical information and supervised machine learning algorithms. The system will be capable of analyzing prescription information, patient-specific risk factors, and possible drug-drug interactions to determine high-risk prescription patterns. The design of the system will involve four key components: data acquisition, preprocessing and feature extraction, predictive modeling, and clinical decision support.

A. Data Acquisition

The system integrates structured medical data from electronic health records (EHRs) [12], prescriptions, and clinical inputs [13]. This includes the demographics of the patients, their age, gender, and weight, as well as their medical history, co-existing conditions, medication, the strength of the medication, the dosage, and the interactions of the medication. All the data used to train and test the model is kept anonymous to ensure the privacy of the patients.

B. Data Preprocessing and Feature Engineering

Medical data may have missing entries, inconsistencies, and categorical variables that need to be properly converted before analysis. Thus, the following preprocessing steps are involved:

- Deletion of duplicate or inconsistent entries.
- Treatment of missing entries using proper imputation methods.
- Conversion of categorical variables into a numerical form.
- Normalization of numerical variables to retain consistency in scales.

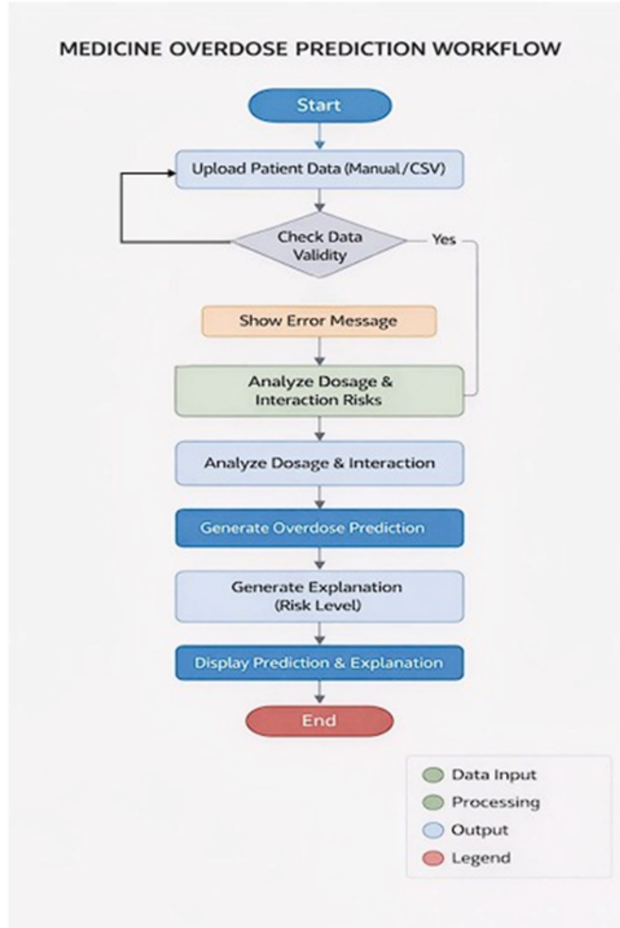

Fig. 4. Design Phase

Feature engineering is then applied to generate clinically meaningful indicators such as total daily dosage, polypharmacy level, and drug interaction severity scores [10]. These engineered features improve the model's ability to detect overdose risks accurately.

C. Machine Learning-Based Overdose Prediction

The overdose prediction task is treated as a supervised classification problem. Machine learning models are trained using historical prescription data labeled as safe or high-risk cases [14]. The following algorithms are implemented and evaluated:

- Decision Tree Classifier.
- Random Forest Classifier [6].
- Artificial Neural Network (ANN) [2].

The Decision Tree model provides interpretability, while Random Forest improves robustness. The ANN captures complex nonlinear relationships among dosage levels and health conditions [3]. The models are trained and validated using performance metrics like accuracy and F1-score [5].

D. Risk Classification and Alert Generation

Depending on the expected outcome, the prescription is categorized into three levels of risk, which include Low Risk,

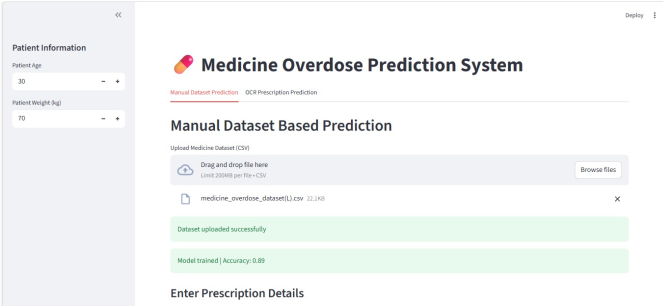

Fig. 5. Medicine overdose prediction

Moderate Risk, and High Risk. In the event that the prescription is classified as high risk, the clinician receives real-time notifications, which include notifications about unsafe doses of medication and potential drug interactions, among other notifications. The system also has a feature that allows for integration with EHR systems.

E. System Workflow

The process, as a whole, involves gathering the patient information, data preprocessing, training of the machine learning model, and identification of the risk level of the overdose. [1].

VI. Experimental Results and Performance Analysis

The part below involves the experimentation of the Medicine Overdose Prediction system, as proposed, and would help in determining the efficiency of the machine learning model in identifying risky prescriptions and the elimination of false alarm cases. [5].

A. Experimental Setup

The experiment would utilize a dataset containing patients' information, categorized as anonymous [13]. This dataset would then be divided into two sets, 80% for the model and 20% for testing. Three machine learning algorithms, namely Decision Tree, Random Forest, and ANN, would be implemented. [14]

B. Performance Metrics

The experiment would use metrics such as accuracy, precision, recall, and F1 score [5]. These metrics are significant for the Medicine Overdose Prediction system to ensure there are no false negatives.

C. Model Performance Comparison

Experimental results show that the Random Forest model performed the best [6].

TABLE II: Classification Performance Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest	89	87	91	89
Decision Tree	84	82	85	83
Artificial Neural Network	88	86	89	87

The Random Forest model showed better recall performance, which is crucial for minimizing the number of missed high-risk prescriptions [6].

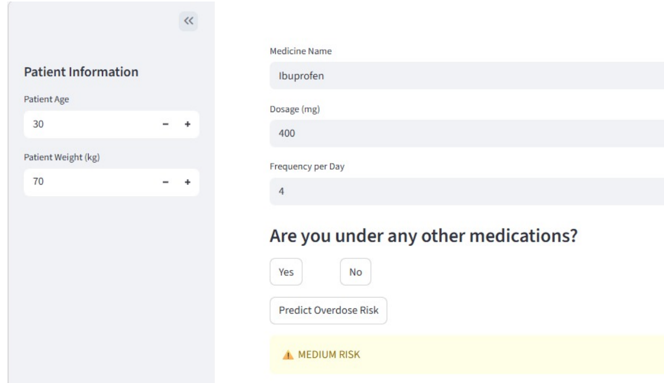

Fig. 6. Patient Information

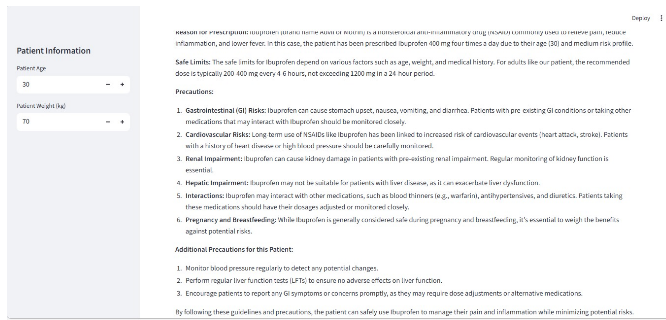

Fig. 7. Descryption of Results

D. OCR Module Performance

The OCR module achieved an average text extraction accuracy of 92% for printed prescriptions [15]. Despite minor extraction inaccuracies, the preprocessing module effectively normalized extracted data.

E. Discussion

The results demonstrate that the system effectively identifies high-risk prescriptions with high reliability [1]. The integration of OCR and ML significantly reduces interpretation errors and improves clinical workflow efficiency [14].

VII. Conclusion and Future Work

This paper outlines the design of the intelligent system that integrates patient information with machine learning techniques [1]. As seen from the experimental results, the predictive capabilities of the intelligent system are impressive, particularly with respect to the recall value. Going forward, [9]the intelligent system could be improved by considering the integration of more sophisticated deep machine learning techniques, including Convolutional Neural Networks and Recurrent Neural Networks [6]. Another way of improving the intelligent system is the integration of the detection of drug-drug interactions with the help of vast pharmaceutical databases [7]. In addition, the personalization component could be improved by considering the age, weight, medical history, and allergies of the patient to better evaluate the possibility of overdosage [8]. Another way of improving the intelligent system is the integration of the natural language processing component to better evaluate the prescriptions. Furthermore, the intelligent system could be improved by developing cloud-based healthcare platforms and mobile health apps. [10] [11]

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